HSE 103 Final Project

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Data Import

```
# Read in covariate data
covarData <- read.csv(file = "GSE85307_covar-1.csv")

# Read in gene expression data
geneExpression <- read.csv(file = "GSE85307_geneExpression.csv",
                           row.names = 1)

# Read in gene name linker
geneNames <- read.csv(file = "GSE85307_geneNames.csv",
                      row.names = 1)

# Check dimensions
dim(covarData)
```

```
## [1] 157  10
```

```
dim(geneExpression)
```

```
## [1] 500 157
```

```
dim(geneNames)
```

```
## [1] 750   1
```

```
# Check that sample IDs match
all(covarData$geoID %in% colnames(geneExpression))
```

```
## [1] TRUE
```

```
all(colnames(geneExpression) %in% covarData$geoID)
```

```
## [1] TRUE
```

Selected Genes

```
# Store selected genes
selectedGenes <- c("TBL1X",
                   "KDM1B",
                   "TMEM176B")

# Identify probe IDs using R
geneProbeMap <- geneNames[
  geneNames$GENE %in% selectedGenes,
  ,
  drop = FALSE
]

# Save probe IDs from row names
geneProbeMap$PROBEID <- rownames(geneProbeMap)

# Put genes in selected order
geneProbeMap <- geneProbeMap[
  match(selectedGenes,
        geneProbeMap$GENE),
  ,
  drop = FALSE
]

# Check probe IDs
geneProbeMap
```

```
##             GENE PROBEID
## 8165911    TBL1X 8165911
## 8117081    KDM1B 8117081
## 8143790 TMEM176B 8143790
```

```
# Verify that all selected genes were found
all(selectedGenes %in%
      geneProbeMap$GENE)
```

```
## [1] TRUE
```

```
# Verify that all probe IDs are in expression data
all(geneProbeMap$PROBEID %in%
      rownames(geneExpression))
```

```
## [1] TRUE
```

Data Wrangling

```
# Create project data frame
projectData <- covarData
```

```r
# Remove text from BMI values
projectData$mat_BMI <- sub(
  "bmi_kg/m2: ",
  "",
  projectData$mat_BMI
)

# Convert BMI to numeric
projectData$mat_BMI <- as.numeric(
  projectData$mat_BMI
)

# Calculate median BMI
medianBMI <- median(
  projectData$mat_BMI,
  na.rm = TRUE
)

# Create categorical BMI variable
# Labels incorporate Submission 1 feedback
projectData$BMIgroup <- cut(
  projectData$mat_BMI,
  breaks = c(-Inf,
             medianBMI,
             Inf),
  labels = c("Below Median BMI",
             "Above Median BMI")
)

# Remove text from vitamin D values
projectData$mat_vitD <- sub(
  "vitamind_25\\(0h\\)_ng/ml_whole_blood_baseline: ",
  "",
  projectData$mat_vitD
)

# Convert vitamin D to numeric
projectData$mat_vitD <- as.numeric(
  projectData$mat_vitD
)

# Remove text from pregnancy outcome values
projectData$preg_preeclampsia <- sub(
  "pregnancy_condition: ",
  "",
  projectData$preg_preeclampsia
)

# Set pregnancy outcome order
projectData$preg_preeclampsia <- factor(
  projectData$preg_preeclampsia,
  levels = c("Normal",
             "Preeclampsia")
```

```
)

# Check cleaned variables
summary(projectData$mat_BMI)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   17.95   22.90   27.33   28.94   33.27   61.95
```

```
summary(projectData$mat_vitD)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    5.11   13.40   20.40   22.98   29.50   80.80
```

```
table(projectData$BMIgroup)
```

```
##
## Below Median BMI Above Median BMI
##               79               78
```

```
table(projectData$preg_preeclampsia)
```

```
##
##       Normal Preeclampsia
##          110           47
```

Add Selected Gene Expression

```
# Select expression rows using probe IDs identified in R
selectedExpression <- geneExpression[
  geneProbeMap$PROBEID,
  projectData$geoID,
  drop = FALSE
]

# Replace probe IDs with gene names
rownames(selectedExpression) <-
  geneProbeMap$GENE

# Transpose so genes become columns
selectedExpression <-
  as.data.frame(
    t(selectedExpression)
  )

# Combine covariate and expression data
analysisData <- cbind(
  projectData,
  selectedExpression
```

```
)

# Check selected gene values
summary(
  analysisData[, selectedGenes]
)
```

```
##      TBL1X           KDM1B          TMEM176B
##  Min.   :8.283   Min.   :7.547   Min.   :7.373
##  1st Qu.:8.955   1st Qu.:8.191   1st Qu.:7.842
##  Median :9.194   Median :8.393   Median :8.278
##  Mean   :9.165   Mean   :8.374   Mean   :8.416
##  3rd Qu.:9.421   3rd Qu.:8.569   3rd Qu.:8.933
##  Max.   :9.862   Max.   :8.955   Max.   :9.959
```

```
# Check categorical variables
table(analysisData$BMIgroup)
```

```
##
## Below Median BMI Above Median BMI
##               79               78
```

```
table(analysisData$preg_preeclampsia)
```

```
##
##       Normal Preeclampsia
##          110           47
```

Plotting Function

```
buildGenePlots <- function(data, geneList) {

  # Create empty list to store plots
  plotList <- list()

  # Loop through each gene
  for (gene in geneList) {

    # Make a temporary copy of the data
    tempData <- data

    # Save current gene expression in one column
    tempData$geneExpression <-
      tempData[, gene]

    # Create italicized gene labels
    geneAxis <- TeX(
      paste0(
        "$\\it{",
```

```
      gene,
      "}$ Gene Expression"
    )
  )

  histTitle <- TeX(
    paste0(
      "$\\it{",
      gene,
      "}$ Gene Expression Distribution"
    )
  )

  scatterTitle <- TeX(
    paste0(
      "Maternal Vitamin D and ",
      "$\\it{",
      gene,
      "}$ Expression"
    )
  )

  boxTitle <- TeX(
    paste0(
      "$\\it{",
      gene,
      "}$ Expression by Pregnancy Outcome and Maternal BMI"
    )
  )

  # Histogram
  histPlot <- ggplot(
    tempData,
    aes(x = geneExpression)
  ) +
    geom_histogram(
      bins = 30
    ) +
    labs(
      title = histTitle,
      x = geneAxis,
      y = "Frequency"
    ) +
    theme_classic()

  # Scatterplot
  scatterPlot <- ggplot(
    tempData,
    aes(
      x = mat_vitD,
      y = geneExpression
    )
  ) +
    geom_point() +
```

```
    geom_smooth(
      method = lm,
      se = FALSE
    ) +
    labs(
      title = scatterTitle,
      x = "Maternal Vitamin D (ng/mL)",
      y = geneAxis
    ) +
    theme_classic()

  # Boxplot
  boxPlot <- ggplot(
    tempData,
    aes(
      x = preg_preeclampsia,
      y = geneExpression,
      fill = BMIgroup
    )
  ) +
    geom_boxplot() +
    labs(
      title = boxTitle,
      x = "Pregnancy Outcome",
      y = geneAxis,
      fill = "Maternal BMI Group"
    ) +
    theme_classic()

  # Save histogram
  plotList[[
    paste0(
      gene,
      "_histogram"
    )
  ]] <- histPlot

  # Save scatterplot
  plotList[[
    paste0(
      gene,
      "_scatterplot"
    )
  ]] <- scatterPlot

  # Save boxplot
  plotList[[
    paste0(
      gene,
      "_boxplot"
    )
  ]] <- boxPlot
}
```

```
  return(plotList)
}
```

Generate the Nine Required Plots

```
# Run function for all three genes
genePlots <- buildGenePlots(
  data = analysisData,
  geneList = selectedGenes
)

# Confirm that 9 plots were created
length(genePlots)
```

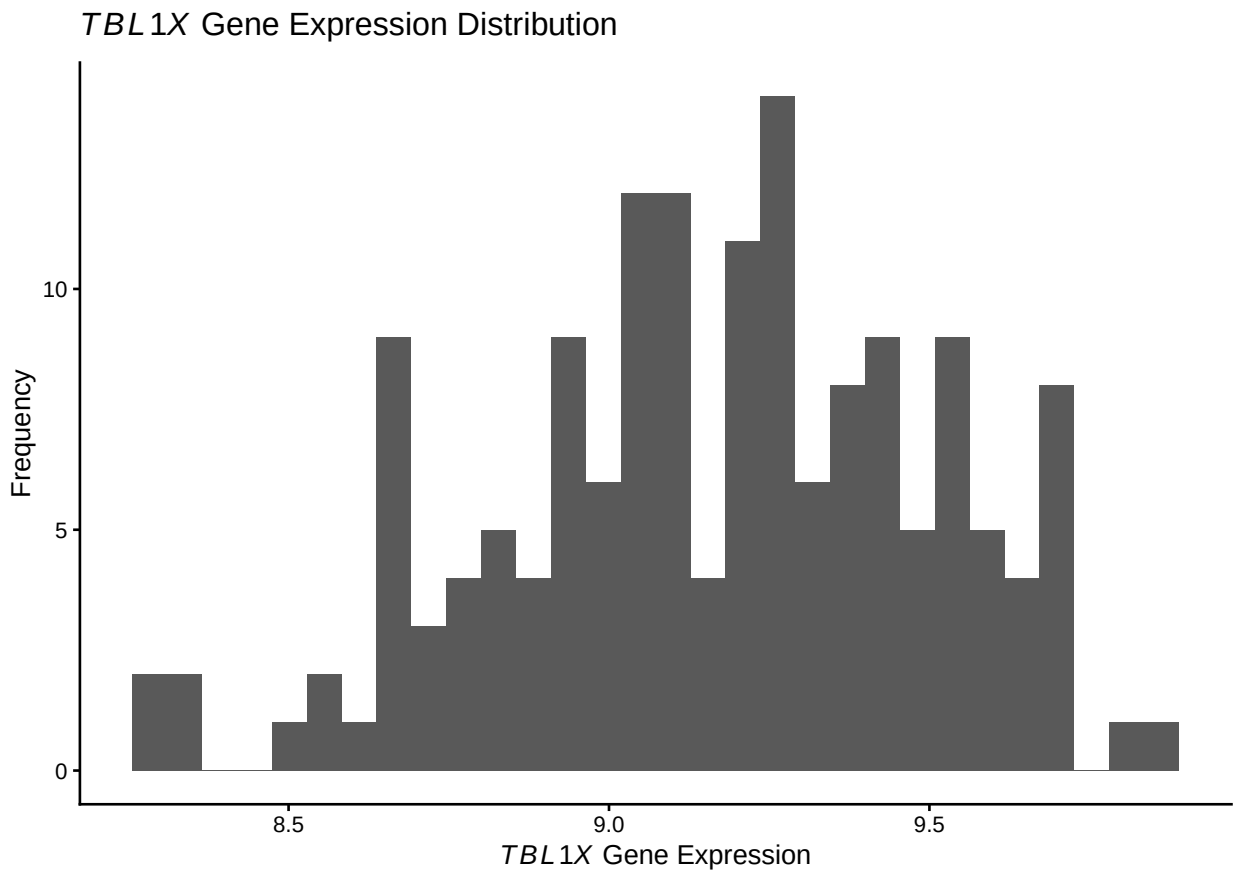
```
## [1] 9
```

```
# Check plot names
names(genePlots)
```

```
## [1] "TBL1X_histogram"      "TBL1X_scatterplot"    "TBL1X_boxplot"
## [4] "KDM1B_histogram"      "KDM1B_scatterplot"    "KDM1B_boxplot"
## [7] "TMEM176B_histogram"   "TMEM176B_scatterplot" "TMEM176B_boxplot"
```

TBL1X

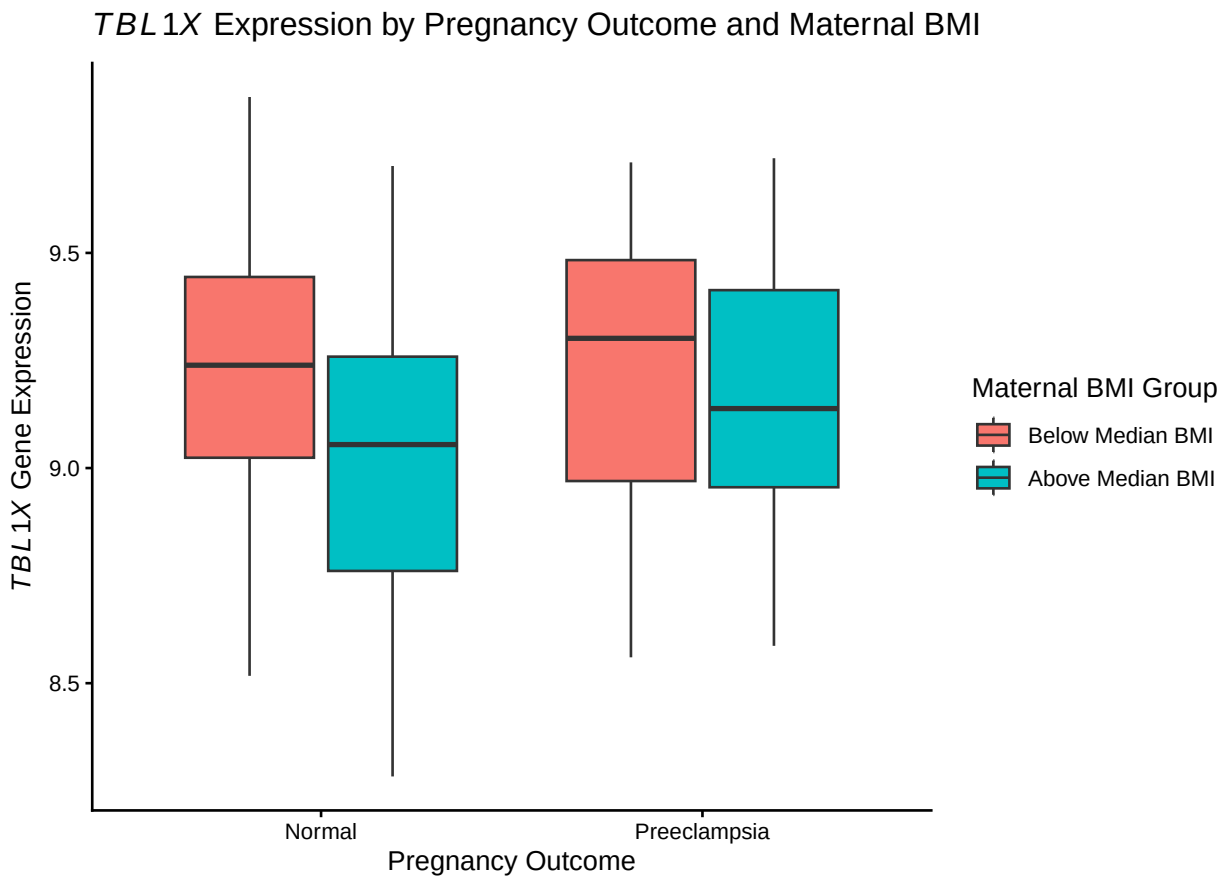
```
genePlots[["TBL1X_histogram"]]
```




```
genePlots[["TBL1X_scatterplot"]]
```

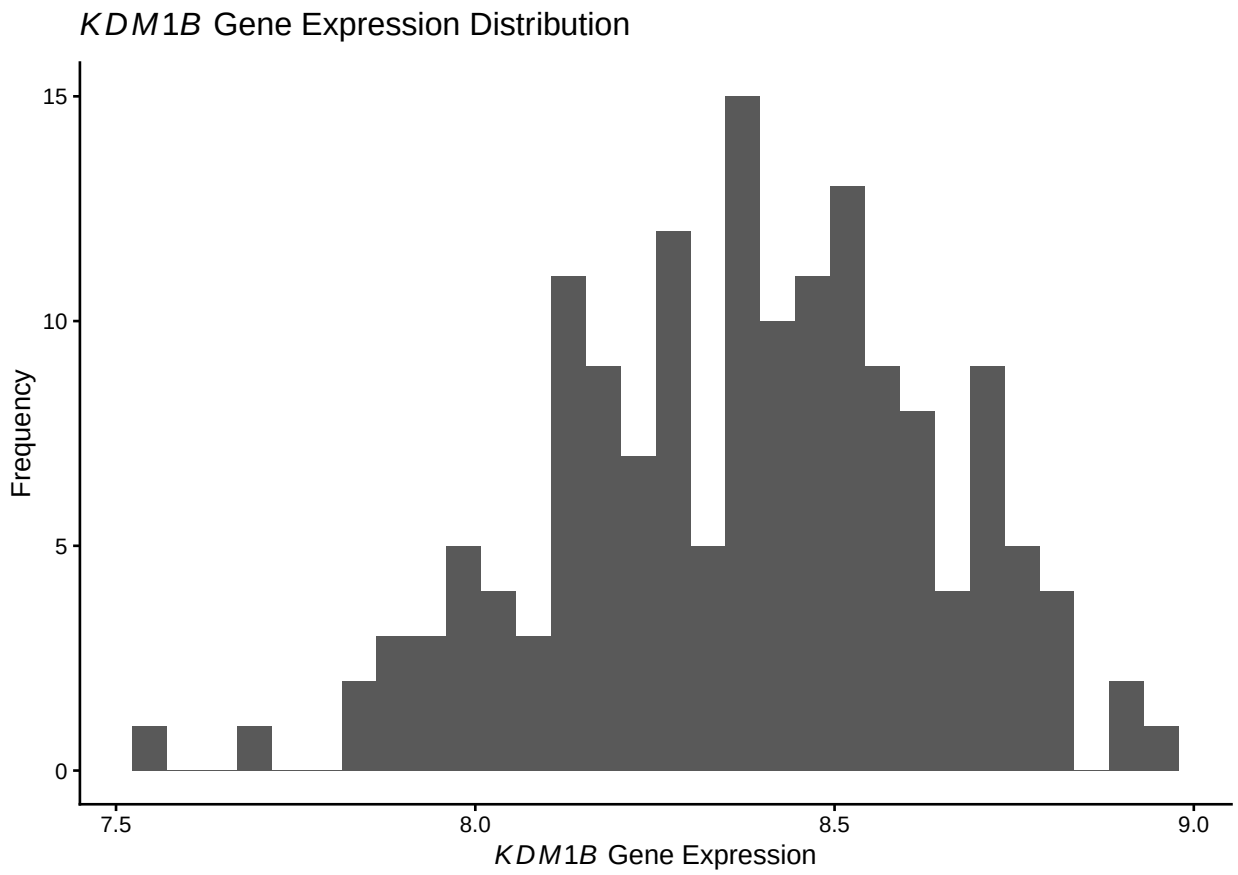



```
genePlots[["TBL1X_boxplot"]]
```

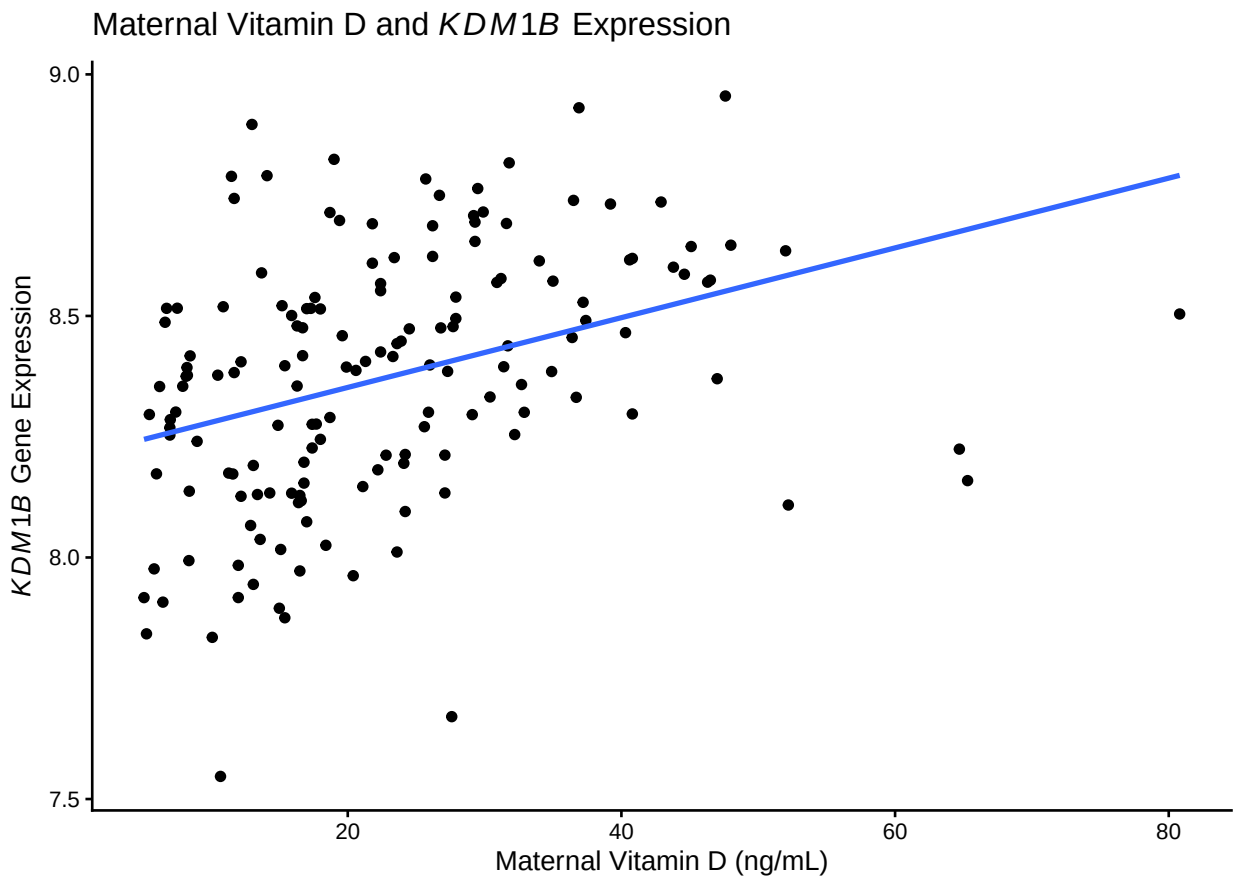


KDM1B

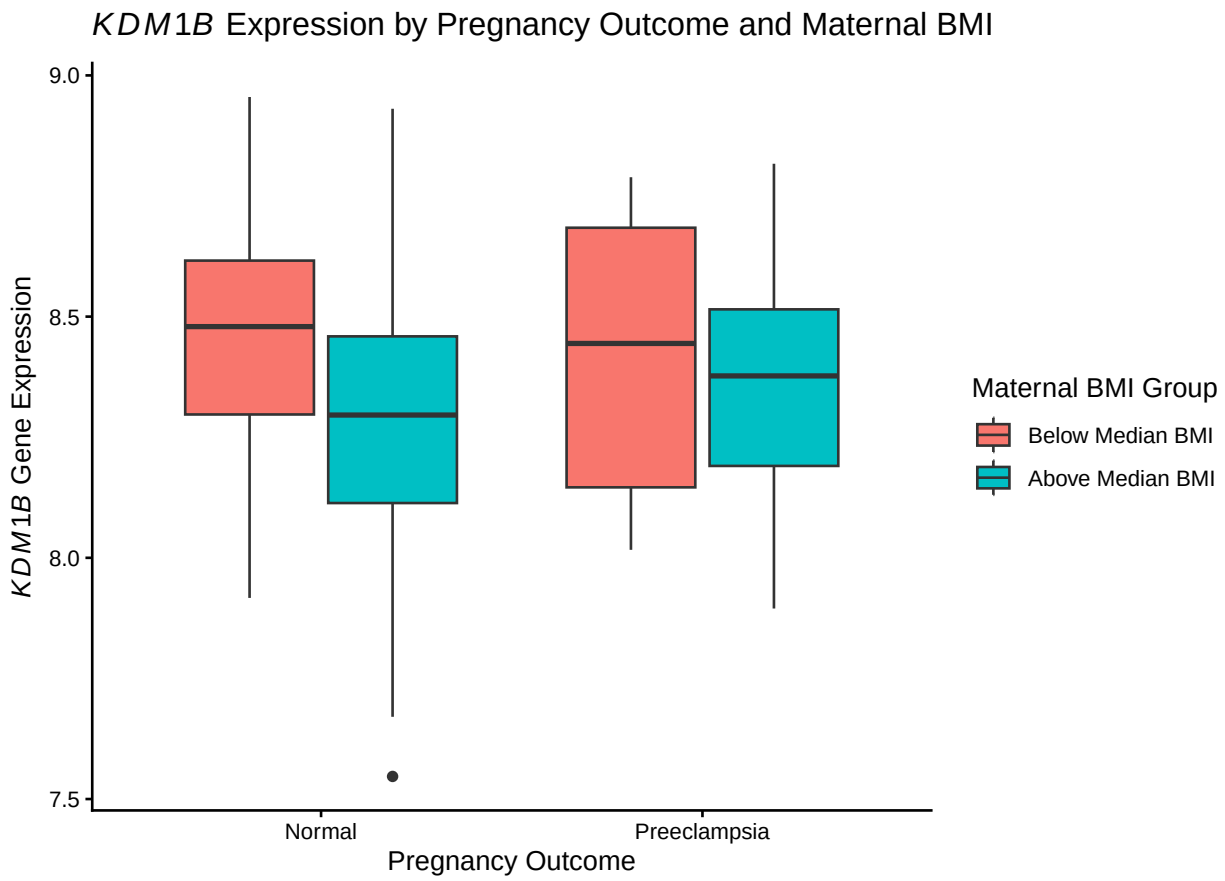
```
genePlots[["KDM1B_histogram"]]
```




```
genePlots[["KDM1B_scatterplot"]]
```

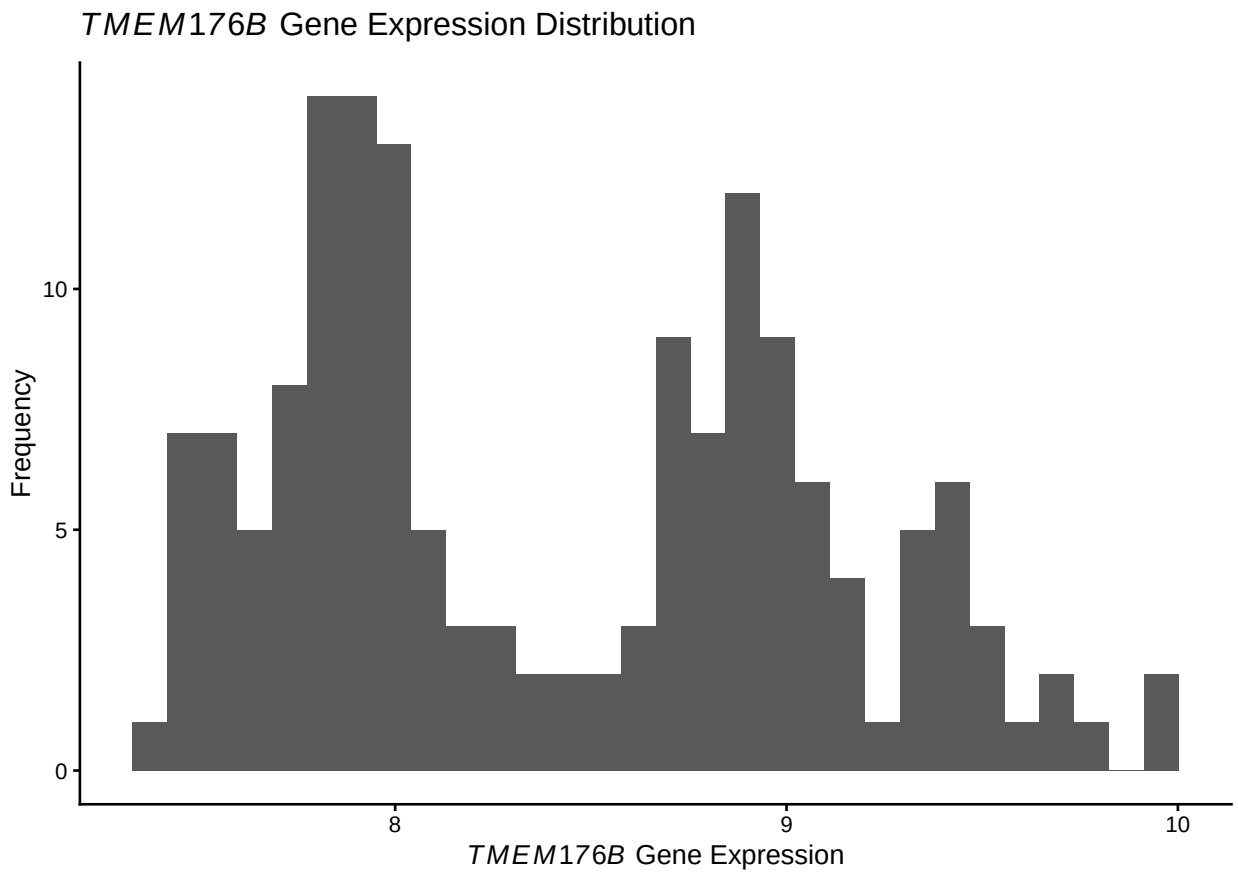



```
genePlots[["KDM1B_boxplot"]]
```

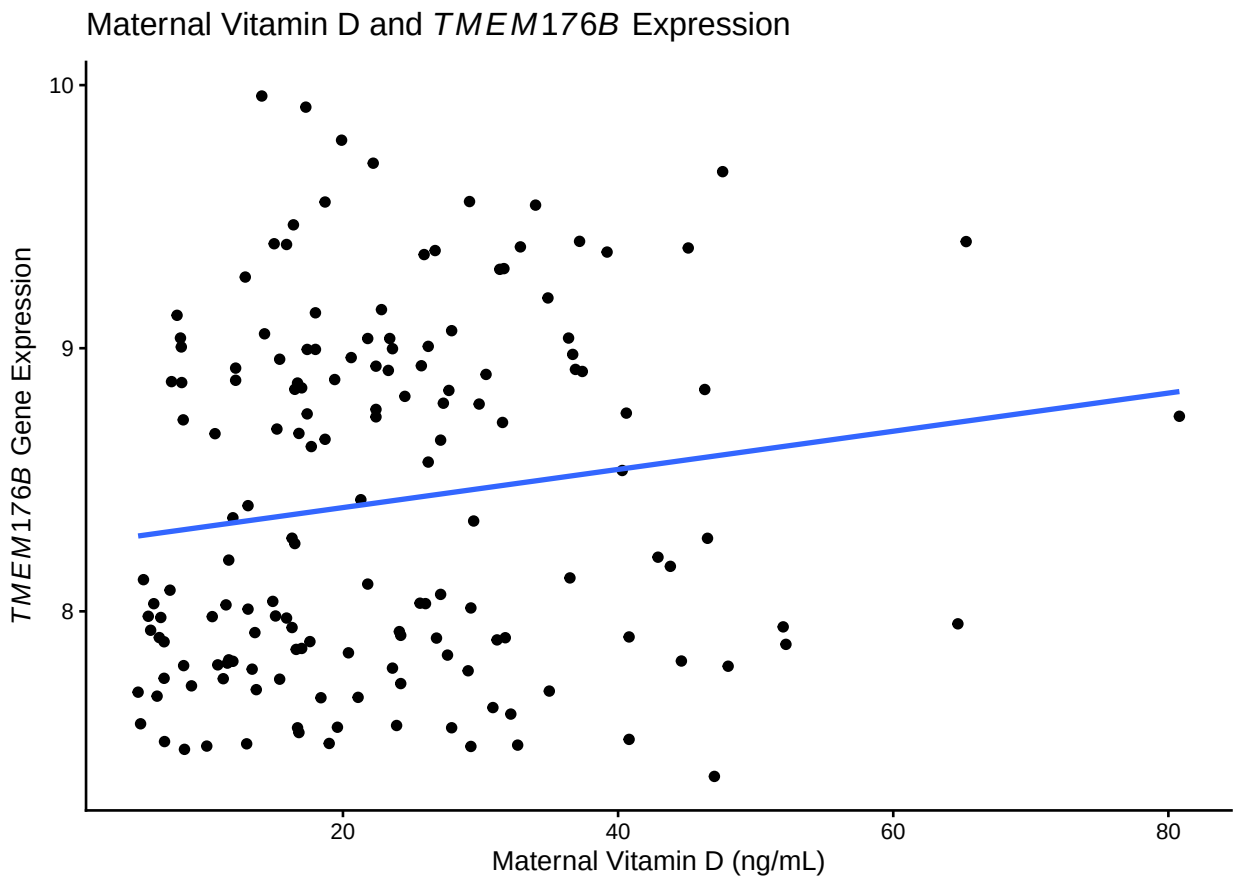


TMEM176B

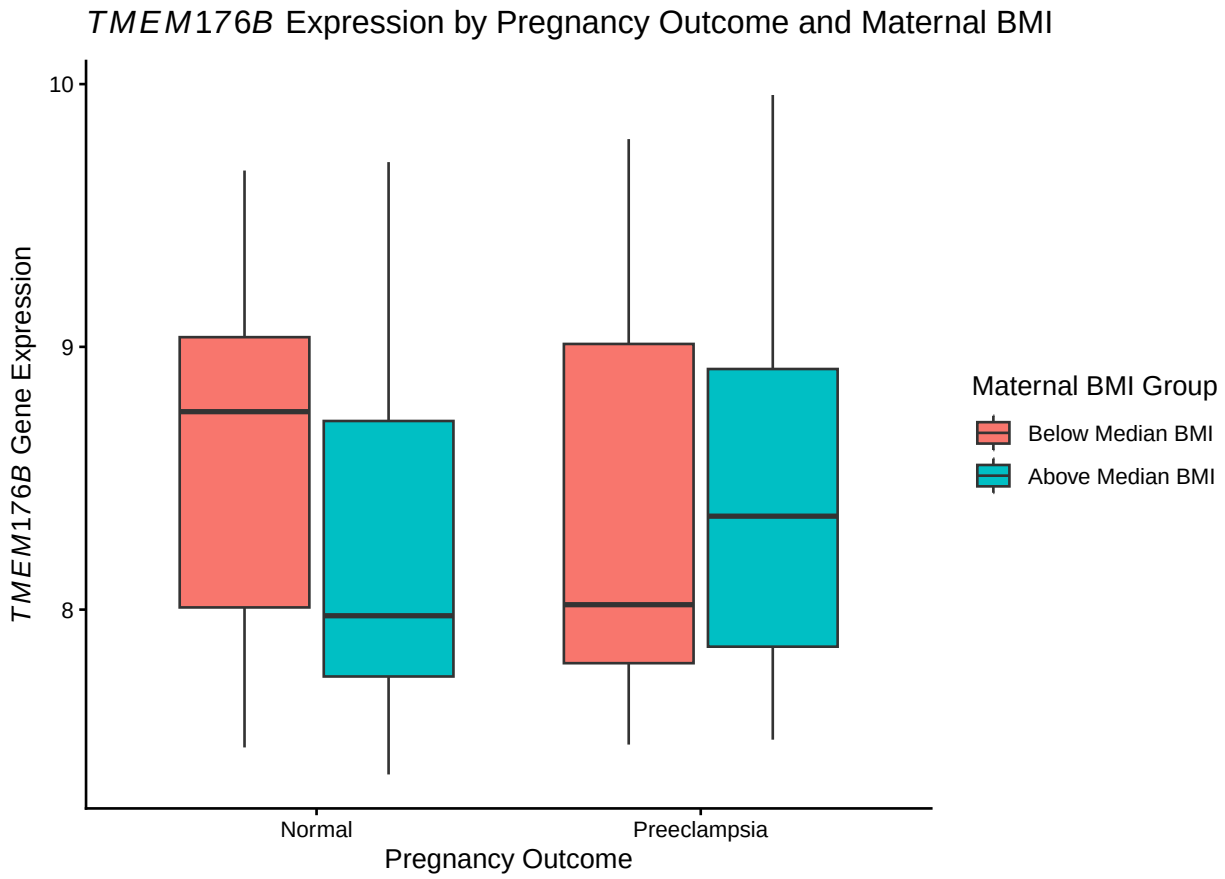
```
genePlots[["TMEM176B_histogram"]]
```




```
genePlots[["TMEM176B_scatterplot"]]
```




```
genePlots[["TMEM176B_boxplot"]]
```



Summary Statistics

```
# Remove text from maternal age values
analysisData$mat_age_yrs <- sub(
  "age_years: ",
  "",
  analysisData$mat_age_yrs
)

# Convert maternal age to numeric
analysisData$mat_age_yrs <- as.numeric(
  analysisData$mat_age_yrs
)

# Check maternal age values
summary(analysisData$mat_age_yrs)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   18.43   22.15   26.57   26.38   30.14   38.76
```

```
# Check maternal asthma values
unique(analysisData$mat_asthma)
```

```
## [1] "asthma_0=healthy_1=asthmatic: 1" "asthma_0=healthy_1=asthmatic: 0"
```

```r
# Remove text from maternal asthma values
analysisData$mat_asthma <- sub(
  "asthma_0=healthy_1=asthmatic: ",
  "",
  analysisData$mat_asthma
)

# Change asthma values to labels
analysisData$mat_asthma <- factor(
  analysisData$mat_asthma,
  levels = c("0", "1"),
  labels = c("Healthy", "Asthmatic")
)

# Check maternal asthma values
table(analysisData$mat_asthma)
```

```
##
##   Healthy Asthmatic
##        94        63
```

Summary Statistics Table

```r
# Separate data by pregnancy outcome
normalData <- analysisData[
  analysisData$preg_preeclampsia == "Normal",
]

preeclampsiaData <- analysisData[
  analysisData$preg_preeclampsia == "Preeclampsia",
]

# Create empty summary table
summaryTable <- data.frame(
  Variable = character(),
  Normal = character(),
  Preeclampsia = character(),
  stringsAsFactors = FALSE
)

# Maternal age
normalMean <- round(mean(normalData$mat_age_yrs,
                         na.rm = TRUE), 2)
normalSD <- round(sd(normalData$mat_age_yrs,
                     na.rm = TRUE), 2)

preeclampsiaMean <- round(mean(preeclampsiaData$mat_age_yrs,
                               na.rm = TRUE), 2)
preeclampsiaSD <- round(sd(preeclampsiaData$mat_age_yrs,
                           na.rm = TRUE), 2)
```

```
summaryTable <- rbind(
  summaryTable,
  data.frame(
    Variable = "Maternal Age, years",
    Normal = paste0(normalMean,
                    " (", normalSD, ")"),
    Preeclampsia = paste0(preeclampsiaMean,
                          " (", preeclampsiaSD, ")")
  )
)

# Maternal BMI
normalMean <- round(mean(normalData$mat_BMI,
                         na.rm = TRUE), 2)
normalSD <- round(sd(normalData$mat_BMI,
                     na.rm = TRUE), 2)

preeclampsiaMean <- round(mean(preeclampsiaData$mat_BMI,
                               na.rm = TRUE), 2)
preeclampsiaSD <- round(sd(preeclampsiaData$mat_BMI,
                           na.rm = TRUE), 2)

summaryTable <- rbind(
  summaryTable,
  data.frame(
    Variable = "Maternal BMI",
    Normal = paste0(normalMean,
                    " (", normalSD, ")"),
    Preeclampsia = paste0(preeclampsiaMean,
                          " (", preeclampsiaSD, ")")
  )
)

# Maternal vitamin D
normalMean <- round(mean(normalData$mat_vitD,
                         na.rm = TRUE), 2)
normalSD <- round(sd(normalData$mat_vitD,
                     na.rm = TRUE), 2)

preeclampsiaMean <- round(mean(preeclampsiaData$mat_vitD,
                               na.rm = TRUE), 2)
preeclampsiaSD <- round(sd(preeclampsiaData$mat_vitD,
                           na.rm = TRUE), 2)

summaryTable <- rbind(
  summaryTable,
  data.frame(
    Variable = "Maternal Vitamin D, ng/mL",
    Normal = paste0(normalMean,
                    " (", normalSD, ")"),
    Preeclampsia = paste0(preeclampsiaMean,
                          " (", preeclampsiaSD, ")")
  )
```

```
)

# BMI group header
summaryTable <- rbind(
  summaryTable,
  data.frame(
    Variable = "Maternal BMI Group",
    Normal = "",
    Preeclampsia = ""
  )
)

# BMI group categories
for (level in levels(analysisData$BMIgroup)) {

  normalN <- sum(normalData$BMIgroup == level,
                 na.rm = TRUE)

  normalPerc <- round(
    normalN / nrow(normalData) * 100,
    1
  )

  preeclampsiaN <- sum(
    preeclampsiaData$BMIgroup == level,
    na.rm = TRUE
  )

  preeclampsiaPerc <- round(
    preeclampsiaN /
      nrow(preeclampsiaData) * 100,
    1
  )

  summaryTable <- rbind(
    summaryTable,
    data.frame(
      Variable = paste0("  ", level),
      Normal = paste0(normalN,
                      " (", normalPerc, "%)"),
      Preeclampsia = paste0(preeclampsiaN,
                            " (", preeclampsiaPerc, "%)")
    )
  )
}

# Maternal asthma header
summaryTable <- rbind(
  summaryTable,
  data.frame(
    Variable = "Maternal Asthma",
    Normal = "",
    Preeclampsia = ""
```

```
  )
)

# Maternal asthma categories
for (level in levels(analysisData$mat_asthma)) {

  normalN <- sum(normalData$mat_asthma == level,
                 na.rm = TRUE)

  normalPerc <- round(
    normalN / nrow(normalData) * 100,
    1
  )

  preeclampsiaN <- sum(
    preeclampsiaData$mat_asthma == level,
    na.rm = TRUE
  )

  preeclampsiaPerc <- round(
    preeclampsiaN /
      nrow(preeclampsiaData) * 100,
    1
  )

  summaryTable <- rbind(
    summaryTable,
    data.frame(
      Variable = paste0("  ", level),
      Normal = paste0(normalN,
                      " (", normalPerc, "%)"),
      Preeclampsia = paste0(preeclampsiaN,
                            " (", preeclampsiaPerc, "%)")
    )
  )
}

# Display summary table
knitr::kable(
  summaryTable,
  caption = "Summary Statistics Stratified by Pregnancy Outcome",
  col.names = c(
    "Variable",
    paste0("Normal (n = ",
           nrow(normalData), ")"),
    paste0("Preeclampsia (n = ",
           nrow(preeclampsiaData), ")")
  ),
  align = c("l", "c", "c")
)
```

Table 1: Summary Statistics Stratified by Pregnancy Outcome

Variable	Normal (n = 110)	Preeclampsia (n = 47)
Maternal Age, years	26.63 (5.08)	25.78 (4.95)
Maternal BMI	27.99 (7.66)	31.16 (8.03)
Maternal Vitamin D, ng/mL	24.37 (14.55)	19.72 (8.33)
Maternal BMI Group		
Below Median BMI	61 (55.5%)	18 (38.3%)
Above Median BMI	49 (44.5%)	29 (61.7%)
Maternal Asthma		
Healthy	67 (60.9%)	27 (57.4%)
Asthmatic	43 (39.1%)	20 (42.6%)

Heatmap

```
# Match gene names to probes in the expression data
heatmapGeneMap <- geneNames[
  rownames(geneNames) %in% rownames(geneExpression),
  ,
  drop = FALSE
]

# Save probe IDs
heatmapGeneMap$PROBEID <- rownames(heatmapGeneMap)

# Remove missing gene names
heatmapGeneMap <- heatmapGeneMap[
  !is.na(heatmapGeneMap$GENE),
  ,
  drop = FALSE
]

# Calculate variance for each probe
geneVariance <- apply(
  geneExpression,
  1,
  var
)

# Add variance to gene map
heatmapGeneMap$Variance <-
  geneVariance[heatmapGeneMap$PROBEID]

# Remove genes already used in Submission 2
additionalGenes <- heatmapGeneMap[
  !(heatmapGeneMap$GENE %in% selectedGenes),
  ,
  drop = FALSE
]

# Put highest variance genes first
```

```
additionalGenes <- additionalGenes[
  order(
    additionalGenes$Variance,
    decreasing = TRUE
  ),
]

# Remove duplicate gene names
additionalGenes <- additionalGenes[
  !duplicated(additionalGenes$GENE),
]

# Select top 7 additional genes
additionalGenes <- additionalGenes[1:7, ]

# Combine with the 3 selected genes
heatmapGenes <- c(
  selectedGenes,
  additionalGenes$GENE
)

# Check selected genes
heatmapGenes
```

```
##  [1] "TBL1X"    "KDM1B"    "TMEM176B" "UTS2"     "RUNDC3A"  "CEACAM6"
##  [7] "CLIC2"    "SLC1A5"   "ZNF594"   "ZBTB41"
```

```
length(heatmapGenes)
```

```
## [1] 10
```

```
# Find probe IDs for the 10 heatmap genes
heatmapMap <- heatmapGeneMap[
  heatmapGeneMap$GENE %in% heatmapGenes,
  ,
  drop = FALSE
]

# Put genes in selected order
heatmapMap <- heatmapMap[
  match(
    heatmapGenes,
    heatmapMap$GENE
  ),
  ,
  drop = FALSE
]

# Create expression matrix
heatmapData <- geneExpression[
  heatmapMap$PROBEID,
  analysisData$geoID,
```

```
  drop = FALSE
]

# Replace probe IDs with gene names
rownames(heatmapData) <- heatmapMap$GENE

# Convert to matrix
heatmapData <- as.matrix(heatmapData)

# Check dimensions
dim(heatmapData)
```

```
## [1]  10 157
```

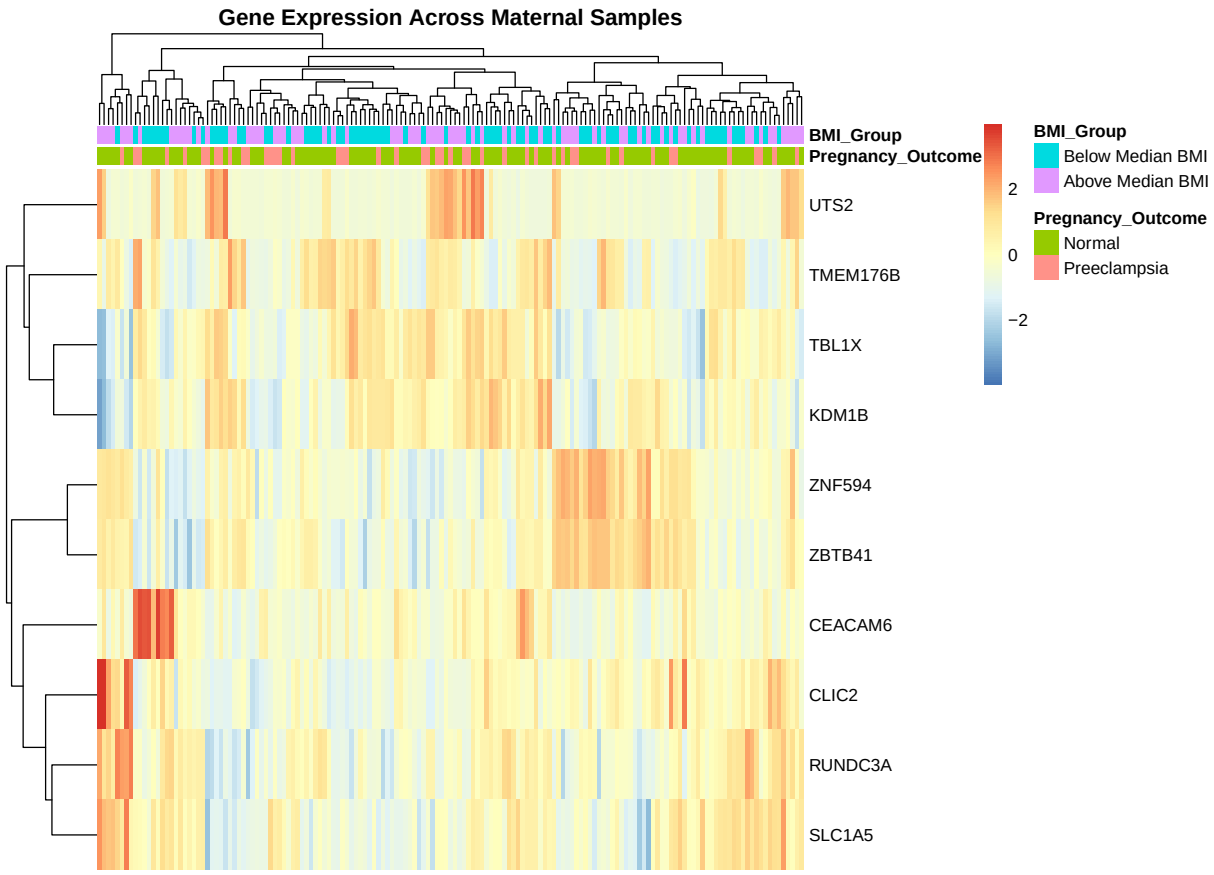
```
# Create tracking bar information
heatmapAnnotation <- data.frame(
  Pregnancy_Outcome = analysisData$preg_preeclampsia,
  BMI_Group = analysisData$BMIgroup
)

# Match annotation rows to heatmap columns
rownames(heatmapAnnotation) <- analysisData$geoID

# Check annotation data
head(heatmapAnnotation)
```

```
##            Pregnancy_Outcome        BMI_Group
## GSM2264346      Preeclampsia Above Median BMI
## GSM2264347            Normal Below Median BMI
## GSM2264348            Normal Below Median BMI
## GSM2264349            Normal Above Median BMI
## GSM2264350      Preeclampsia Below Median BMI
## GSM2264351            Normal Below Median BMI
```

```
# Create clustered heatmap
pheatmap(
  heatmapData,
  scale = "row",
  annotation_col = heatmapAnnotation,
  clustering_distance_rows = "euclidean",
  clustering_distance_cols = "euclidean",
  cluster_rows = TRUE,
  cluster_cols = TRUE,
  show_colnames = FALSE,
  main = "Gene Expression Across Maternal Samples"
)
```



Additional Plot

```
# Create empty data frame for mean differences
lollipopData <- data.frame(
  Gene = character(),
  Difference = numeric(),
  stringsAsFactors = FALSE
)

# Identify normal and preeclampsia samples
normalSamples <- analysisData$geoID[
  analysisData$preg_preeclampsia == "Normal"
]

preeclampsiaSamples <- analysisData$geoID[
  analysisData$preg_preeclampsia == "Preeclampsia"
]

# Calculate mean expression difference for each gene
for (gene in heatmapGenes) {

  normalMean <- mean(
    heatmapData[
```

```
      gene,
      normalSamples
    ],
    na.rm = TRUE
  )

  preeclampsiaMean <- mean(
    heatmapData[
      gene,
      preeclampsiaSamples
    ],
    na.rm = TRUE
  )

  meanDifference <- preeclampsiaMean - normalMean

  lollipopData <- rbind(
    lollipopData,
    data.frame(
      Gene = gene,
      Difference = meanDifference
    )
  )
}

# Order genes by mean difference
lollipopData <- lollipopData[
  order(lollipopData$Difference),
]

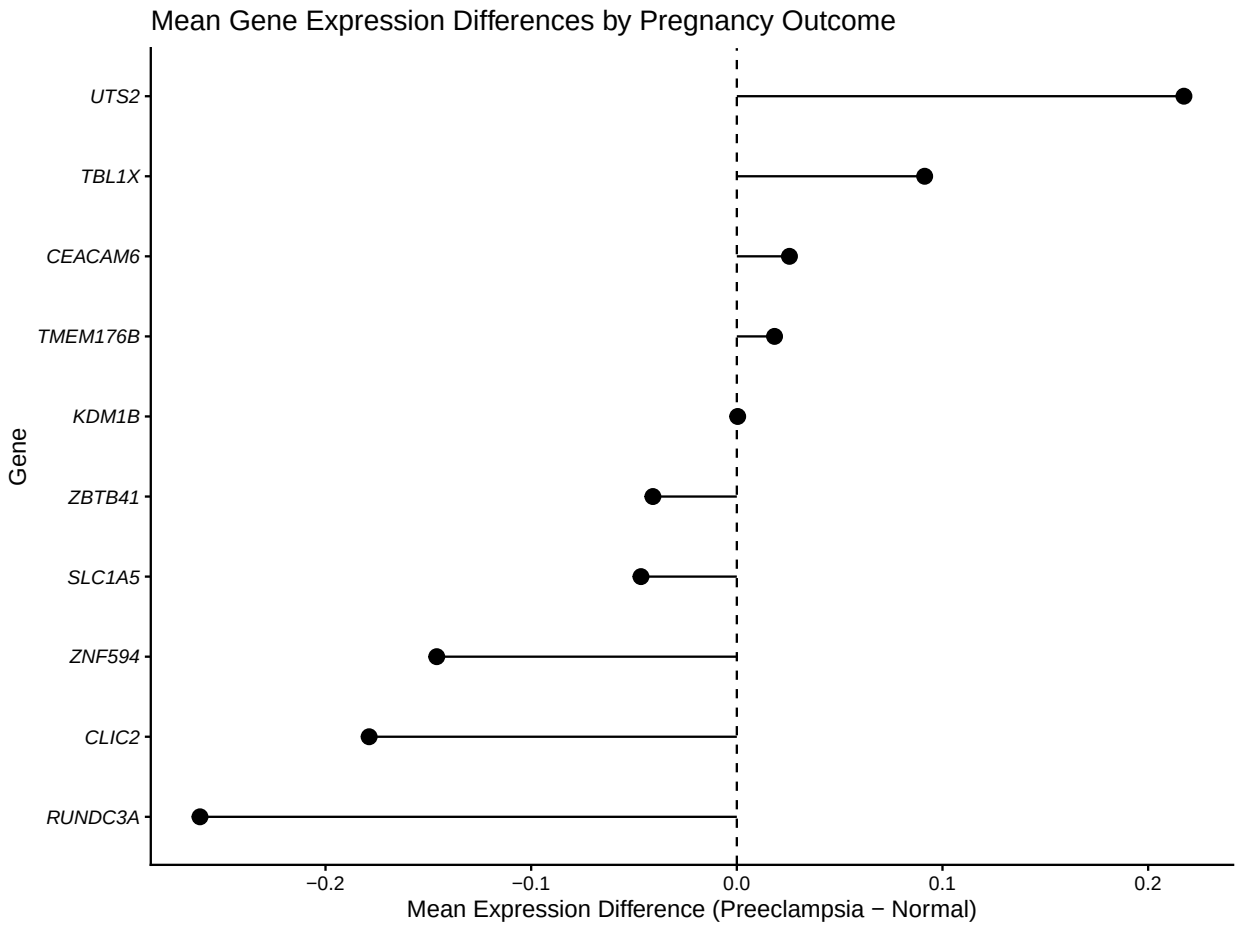
# Set gene order
lollipopData$Gene <- factor(
  lollipopData$Gene,
  levels = lollipopData$Gene
)

# Check data
lollipopData
```

```
##        Gene    Difference
## 5   RUNDC3A -0.2610319660
## 7     CLIC2 -0.1787837116
## 9    ZNF594 -0.1459213600
## 8    SLC1A5 -0.0466180956
## 10   ZBTB41 -0.0408161008
## 2     KDM1B  0.0004152805
## 3  TMEM176B  0.0183335921
## 6   CEACAM6  0.0255862675
## 1     TBL1X  0.0913335170
## 4      UTS2  0.2173926162
```

```
# Create lollipop plot
ggplot(
```

```
  lollipopData,
  aes(
    x = Difference,
    y = Gene
  )
) +
  geom_segment(
    aes(
      x = 0,
      xend = Difference,
      y = Gene,
      yend = Gene
    )
  ) +
  geom_point(
    size = 3
  ) +
  geom_vline(
    xintercept = 0,
    linetype = "dashed"
  ) +
  labs(
    title = "Mean Gene Expression Differences by Pregnancy Outcome",
    x = "Mean Expression Difference (Preeclampsia - Normal)",
    y = "Gene"
  ) +
  scale_y_discrete(
    labels = function(x) {
      parse(
        text = paste0(
          "italic('",
          x,
          "')"
        )
      )
    }
  ) +
  theme_classic()
```


Mean Gene Expression Differences by Pregnancy Outcome
UTS2
TBL1X
CEACAM6
TMEM176B
KDM1B
ZBTB41
SLC1A5
ZNF594
CLIC2
RUNDC3A
Gene
−0.2
−0.1
0.0
0.1
0.2
Mean Expression Difference (Preeclampsia − Normal)